F5: ROBDD size of the decomposable set X_2 and the Legendre set X_E against random sets

Run F5-RUN-01 of study `compiled_advice_robdd_20261110`. Class: boundary diagnostic; no S, no rho row, no leaderboard change. All measured numbers below are `[pcc:unverified]` (no pcc claim id exists for this run yet).

Question

Is the set of decomposable abscissae X_2 (or the Legendre set X_E) more compressible as a reduced ordered BDD than a uniformly random set of the same size, by the margin frozen in PROTOCOL.md (ratio ≤ 0.9, decreasing in n)? The preprocessing bound $S \cdot T^2 \gtrsim n$ charges advice by its size; it does not forbid advice that is more compressible than random with a fast membership query (RESEARCH_PRIME_FIELD_BREAKTHROUGH_PROGRAM.md §1; idea E1).

Mechanism

The indicator of each set, as a function of the n bits of x, is built bottom-up as an ROBDD with a unique table. Its node count is the size of a small-advice representation; if X_2 had structure the count would fall below that of random sets of equal density.

Method

p = largest prime below 2^n, $n \in \{16, 18, 20, 22, 24\}$, one prime-order curve per n. F = small-x points with $x \le B = \lceil p^{(1/3)} \rceil$; X_2 = signed two-sums of F; X_E = abscissae with Legendre symbol 1 (density $\approx 1/2$); INT = interval $[0, |X_2|)$ as positive control. Orders: msb, lsb, and one budgeted Rudell sifting pass from each (sift_msb, sift_lsb; $2n^2$ swaps, 1.2× growth abort). Null: RAND(X) = 10 uniformly random subsets of [0, p) with |RAND| = |X|, seeds `F5/rand/<n>/<set>/<j>`, same instrument and verification. Every ROBDD is verified by re-enumeration. Native Rust tool, local host (4 vCPU), exact counts, no timing taken.

Results `[pcc:unverified]`

Ratio = nodes(X) / mean nodes(RAND); relsd = sd/mean of the 10 RAND counts. Node counts are listed in the order msb / lsb / sift_msb / sift_lsb.

n	p	set	|X|	nodes msb / lsb / sift_msb / sift_lsb	ratio to RAND mean (relsd), same order	INT nodes (4 orders)
16	65521	X2	400	892 / 834 / 872 / 831	1.029 (0.012) / 1.010 (0.006) / 1.000 (0.011) / 1.007 (0.009)	12 / 12 / 12 / 12
16	65521	X_E	32775	8307 / 8290 / 8311 / 8284	1.001 (0.001) / 1.001 (0.001) / 1.001 (0.001) / 1.000 (0.001)	
18	262139	X2	1018	2003 / 1942 / 2010 / 1943	0.994 (0.012) / 1.003 (0.010) / 0.996 (0.013) / 1.001 (0.007)	17 / 17 / 17 / 17
18	262139	X_E	130988	31052 / 31031 / 31102 / 30992	0.998 (0.001) / 1.001 (0.001) / 1.000 (0.001) / 0.999 (0.001)	
20	1048573	X2	1521	3510 / 3348 / 3482 / 3352	1.023 (0.007) / 1.005 (0.007) / 1.014 (0.010) / 1.005 (0.005)	20 / 20 / 20 / 20
20	1048573	X_E	524784	107055 / 106938 / 107202 / 106863	1.000 (0.001) / 1.000 (0.000) / 1.001 (0.000) / 1.000 (0.000)	
22	4194301	X2	5767	10288 / 10137 / 10293 / 10138	1.001 (0.006) / 1.002 (0.003) / 1.000 (0.006) / 1.005 (0.002)	22 / 22 / 22 / 22
22	4194301	X_E	2097809	326495 / 326410 / 326428 / 326403	1.000 (0.000) / 1.000 (0.000) / 1.000 (0.000) / 1.000 (0.000)	
24	16777213	X2	17411	28158 / 27794 / 28263 / 27905	0.997 (0.003) / 0.999 (0.002) / 1.003 (0.004) / 1.003 (0.003)	24 / 24 / 24 / 24
24	16777213	X_E	8387895	1114066 / 1114057 / 1114064 / 1114055	1.000 (0.000) / 1.000 (0.000) / 1.000 (0.000) / 1.000 (0.000)	

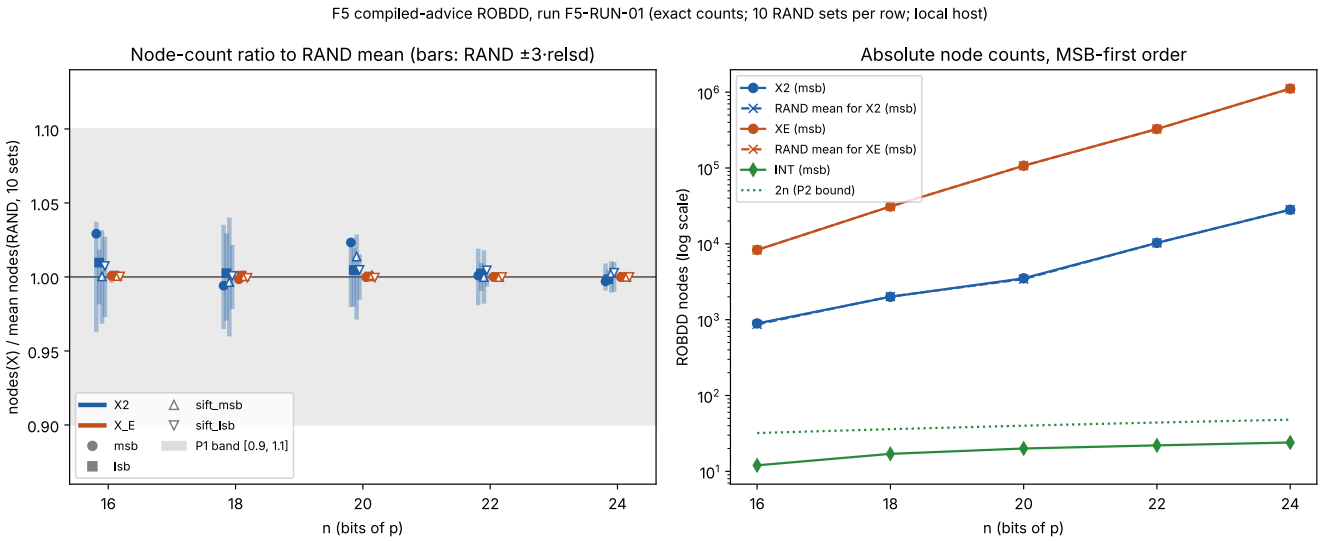


Figure 1: Left: node-count ratio X/mean(RAND) vs n for X_2 and X_E, four orders as markers (open = sifted), bars = RAND ±3·relsd, grey = P1 band [0.9, 1.1]. Right: absolute node counts (log scale), msb order, with RAND means, INT and the 2n bound. Run F5-RUN-01, 10 RAND sets per row, exact counts, local host. Source: `results/run.json`; `figures/make_figure.py`.

Extremes: X_2 ratio 0.994 to 1.029; X_E ratio 0.998 to 1.001 (all 40 cells in [0.9, 1.1]). Largest z-scores: +3.44 (n = 20, X_2, msb), +2.37, +2.35, −2.34; no multiplicity correction preregistered. Costs: 346,541,069 unique-table operations and 21,415 sifting swaps over the X and INT builds; peak RSS 610,432 kB. `[pcc:unverified]`

Reading (PROTOCOL.md §4)

1. Instrument failure: not triggered. INT = 12, 17, 20, 22, 24 nodes against 2n = 32, 36, 40, 44, 48; all 60 ROBDD verifications passed. `[pcc:unverified]`
2. Lead alive: not met; no ratio ≤ 0.9 (minimum 0.994). `[pcc:unverified]`
3. Rule 3 applies, verbatim: "Closed: otherwise: the decomposable and Legendre sets are as incompressible as random sets under ROBDDs at these sizes. Says nothing about other compressed representations." `[pcc:unverified]`

P1 (ratio in [0.9, 1.1] at every n and order, X_2 and X_E): held. P2 (INT ≤ 2n): held. The prior of 0.003 on a downward, decreasing trend was not realised. `[pcc:unverified]`

Scope

Says nothing about other compressed representations, X_3, or n > 24. The §9a isolab-eks confirmatory run is open; the host was shared; no timing is a result. The research-kb server was unavailable; no novelty claim is made.

Provenance

Card	`CARD-compiled-advice-robdd-20261110-F5-RUN-01`, sha256 `e946af9134b86caed6dd735ad8904f262f88cb65f0aae12825de425357fe4a79`
Commit	`da7ac2dbb8dc920652be41efb0be3ee14eaa17a7` (da7ac2dbb8)
Receipt	sha256 `dd28fdb8f727c92a644ce34878d92c0d3b33860c1595e455c26f74a51bdeed05`; status completed; tree dirty before and after (card file, other studies)
Binary	sha256 `7bb1a710a6ae3d3df15a78cde927275195f5cf3c521337d058c6bcbf339f8bb2`
run.json	sha256 `1ab1b4fded683de2fd55b9e53e5f3830a47b9a7229d2f5b2c99d6e18631f0f9a`
FREEZE.sha256	PROTOCOL.md `e57b8ef0449a0353244770e439fc1c1ee7798f9a915e0d8d3cf49513fe4f76d1`; tool/src/main.rs `c0f41c930679fc016f29864e541bac351272dd49976dd8a23fd2a579e015d019`; tool/Cargo.toml `01cec37983a65158e913ab81b4bc6d8ede694b1fb4050e526f6723d243155278`; tool/Cargo.lock `2912a6c720234c63c2c8ed0622fc99d8ff566ee0b80f873b8df3e8aea0c08c29`
Report build	typst Python wheel 0.15.2 (scratch venv); figure by matplotlib 3.11.2